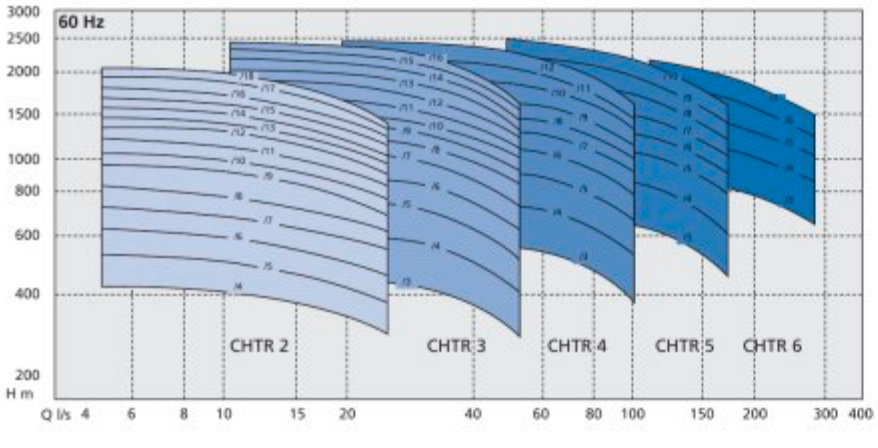
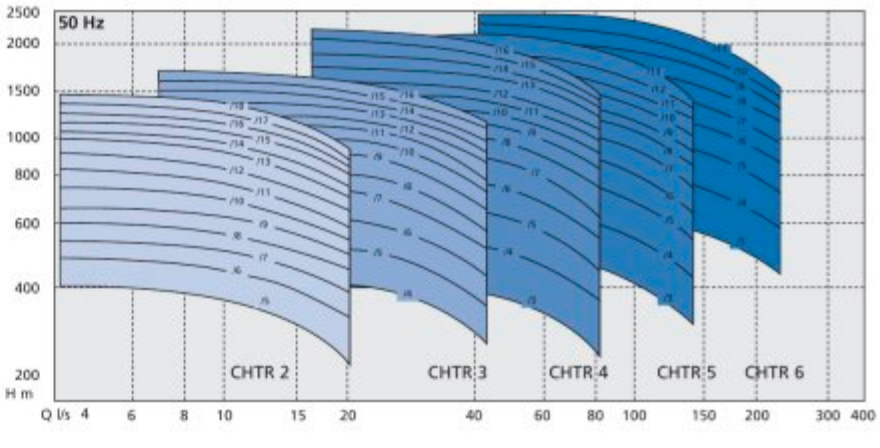
Technical data and tables

Characteristic curves





Technical data	
Flow rate Q	Up to 1,450 m^3/h (400 l/s)
Head H*	Up to 4,000 m
Temperature t	– 60 °C to 450 °C
Pump discharge pressure p_d	Up to 400 bar
Speed	Up to 7,000 rpm
Sizes	DN 50 to DN 250
*Q = 0 and ρ = 1 kg/dm^3	

Materials					
API 610/10 class major components	**S – 5**	**S – 6**	**C – 6**	**A – 8 #**	**D – 1**
Barrel/stage casings	C steel		Cr Ni steel	AUS SS	Duplex
Impellers/diffusers	Cr steel			AUS SS	Duplex
Impeller and stage casing wear rings/bushes	Cr steel			AUS SS*	Duplex*
Shaft	Cr steel/Cr Ni steel			AUS SS*	Duplex*
Shaft sleeve (mech. seal)	AUS SS*				Duplex*

* Hard-faced
On special request

C steel	Carbon steel
Cr steel	Chromium steel
Cr Ni steel	Chromium Nickel steel
AUS SS	Austenitic stainless steel
Duplex	Duplex stainless steel

Applications

Water injection



Refineries



Petrochemical industries



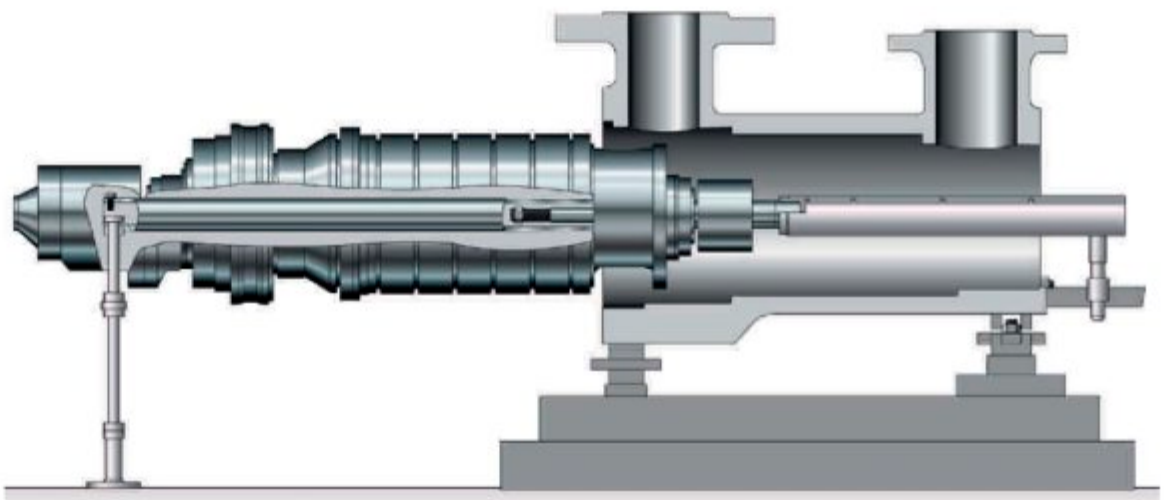
Spectrum

Design	Horizontal, radially split, multistage barrel-type pump, "quick pull-out" cartridge design								
Drive	Electric motor, steam turbine, combustion engine, gas turbine								
Rotation	Clockwise viewed from drive end (suction side drive)								
Nozzles	Flanges to ASME, radial, pointing vertically upwards, to DIN/BS on request								
Mounting	Shaft centreline								
Casing sealing	Confined spiral-wound stainless steel gaskets								
Impellers	Radial flow, single entry, integrally cast								
Pump size		2		3		4		5	
Pump discharge pressure p_d		100	250	100	250	100	250	100	250
Suction-stage impellers	Single entry	▪	▪	▪	▪	▪	▪	▪*	▪
	Double entry					▪	▪	▪*	▪
Axial thrust balancing	Hydraulically by single balance drum	▪	X	▪	X	▪	X	▪	X
	Stepped balance drum	X	▪	X	▪	X	▪	X	▪
Bearings, lubrication									
Radial bearings	Antifriction bearings, splash-lubricated	▪	X	▪	X	▪	X	▪	X
	Plain bearings, forced-feed oil-lubricated	X	▪	X	▪	X	▪	▪	▪
Thrust bearing	Antifriction bearings, splash-lubricated	▪		▪		▪		▪	
	Segmental thrust bearing, Forced-feed oil-lubricated	X	▪	X	▪	X	▪	▪	▪
Shaft seal	Mechanical seal to API 682								
Mechanical seal	Mechanical seal (Cartridge design)	▪	▪	▪	▪	▪	▪	▪	▪

x Not applicable ▪ Applicable * On request

Quick pull-out of the complete cartridge unit is possible, only one screw joint must be undone. The complete cartridge can then be removed easily by means of a special device. The CHTR pump casing remains in the piping system. Therefore, suction and discharge flanges do not need to be disconnected.

The cartridge design of CHTR barrel pumps allows dismantling and replacement of the inner assembly within only a few hours.



Steam generators





Seawater

